Causal Analysis Report

BPI_Challenge_2013_closed_problems.xes.gz-imf

Decision points:	9441
Choice sets:	6
Transitions fitted:	8
Significant (p<0.05):	0
Generated:	2026-06-23 16:11

Step 1 — Association test (logistic regression + permutation)
H□: choice □ util | choice_set.   reject_H0=True → util predicts routing.

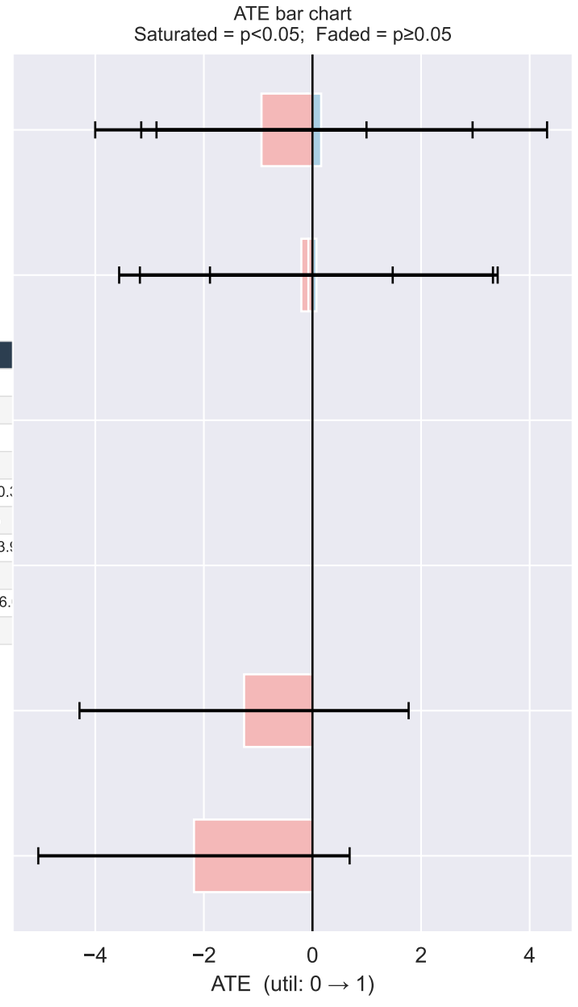
(no data)

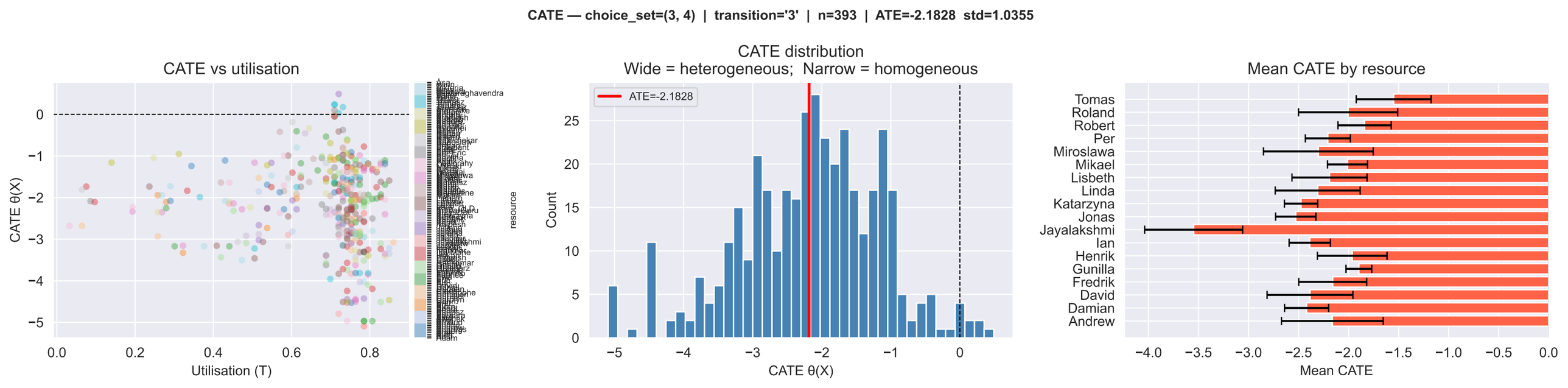
Step 2 — Double ML (partially linear regression, HC3 robust SE)
θ = linear effect of 1-unit ↑ util on P(chose k), after partialling out X.
Equals ATE when the effect is homogeneous. Otherwise: variance-weighted average of CATEs.

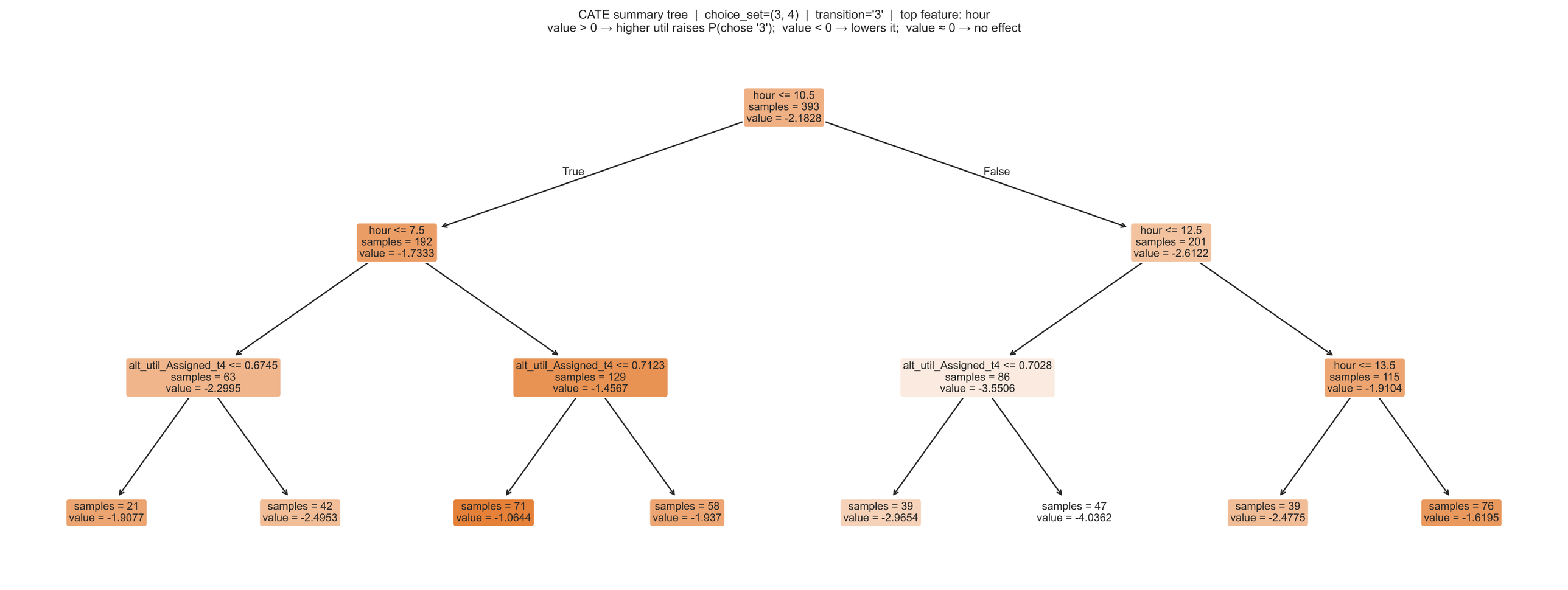
choice_set	choice_set_labels	n	choices_seen	transition	transition_label	alt_utils	theta	se	ci_95	p_value	reject_H0	note
(0, 1, 2)	('т_t0', 'т_t1', 'Unmatched_t2')	4	2	—	—	—	nan	nan	—	nan	—	too few observations
(3, 4)	('Wait_t3', 'Assigned_t4')	393	3, 4	3	Wait_t3	Assigned_t4	-0.729464	0.653073	[-2.0095, 0.5506]	0.264	False	θ=-0.7295: 1-unit ↑ util ↓ P(t3/Wait_t3) by 72.9%
(3, 4)	('Wait_t3', 'Assigned_t4')	393	3, 4	4	Assigned_t4	Wait_t3	-0.689005	0.577212	[-1.8203, 0.4423]	0.2326	False	θ=-0.6890: 1-unit ↑ util ↓ P(t4/Assigned_t4) by 68.9%
(5, 6, 7, 8)	('т_t5', 'т_t6', 'In Progress_t7', 'Awaiting Assignment_t8')	203	7, 8	7	In Progress_t7	Awaiting Assignment_t8	-0.211037	0.352778	[-0.9025, 0.4804]	0.5497	False	θ=-0.2110: 1-unit ↑ util ↓ P(t7/In Progress_t7) by 21.1%
(5, 6, 7, 8)	('т_t5', 'т_t6', 'In Progress_t7', 'Awaiting Assignment_t8')	203	7, 8	8	Awaiting Assignment_t8	In Progress_t7	0.634139	0.470401	[-0.2878, 1.5561]	0.1776	False	θ=+0.6341: 1-unit ↑ util ↑ P(t8/Awaiting Assignment_t8) by 63.4%
(6, 7, 8)	('т_t6', 'In Progress_t7', 'Awaiting Assignment_t8')	1243	7, 8	7	In Progress_t7	Awaiting Assignment_t8	-0.295064	0.15437	[-0.5976, 0.0075]	0.056	False	θ=-0.2951: 1-unit ↑ util ↓ P(t7/In Progress_t7) by 29.5%
(6, 7, 8)	('т_t6', 'In Progress_t7', 'Awaiting Assignment_t8')	1243	7, 8	8	Awaiting Assignment_t8	In Progress_t7	-0.455068	0.19886	[-0.8448, -0.0653]	0.0221	True	θ=-0.4551: 1-unit ↑ util ↓ P(t8/Awaiting Assignment_t8) by 45.5%
(7, 8)	('In Progress_t7', 'Awaiting Assignment_t8')	2239	7, 8	7	In Progress_t7	Awaiting Assignment_t8	0.274858	0.277095	[-0.2682, 0.8180]	0.3212	False	θ=+0.2749: 1-unit ↑ util ↑ P(t7/In Progress_t7) by 27.5%
(7, 8)	('In Progress_t7', 'Awaiting Assignment_t8')	2239	7, 8	8	Awaiting Assignment_t8	In Progress_t7	0.574239	0.331536	[-0.0756, 1.2240]	0.0833	False	θ=+0.5742: 1-unit ↑ util ↑ P(t8/Awaiting Assignment_t8) by 57.4%
(9, 10)	('т_t9', 'Closed_t10')	1446	10	—	—	—	nan	nan	—	nan	—	too few observations

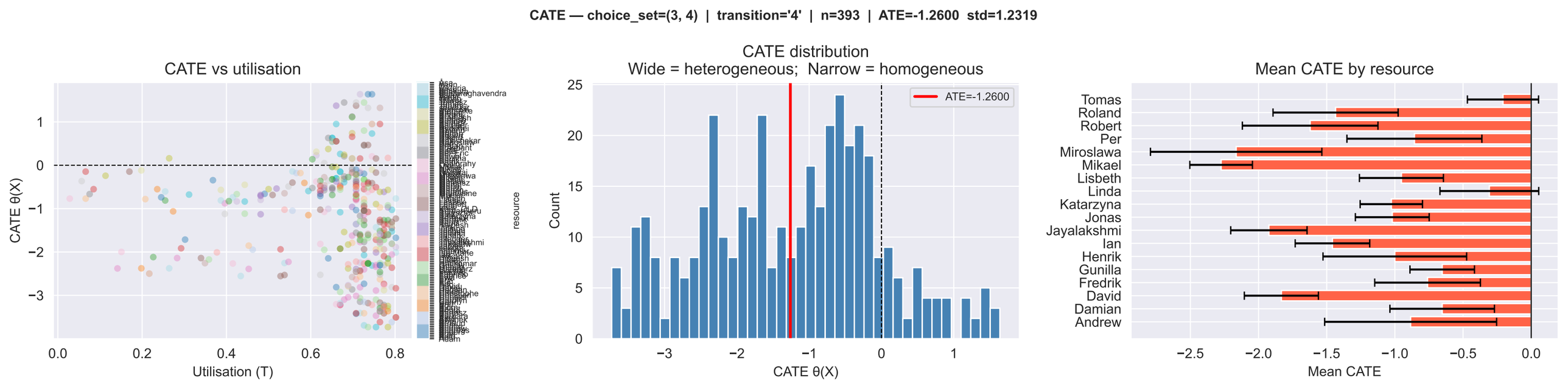
Step 3 — Causal Forest DML (ATE per transition)
ATE = mean of per-sample CATEs. Green rows = H0 rejected (p<0.05).
Less powerful for significance than Double ML, but basis for heterogeneity analysis.

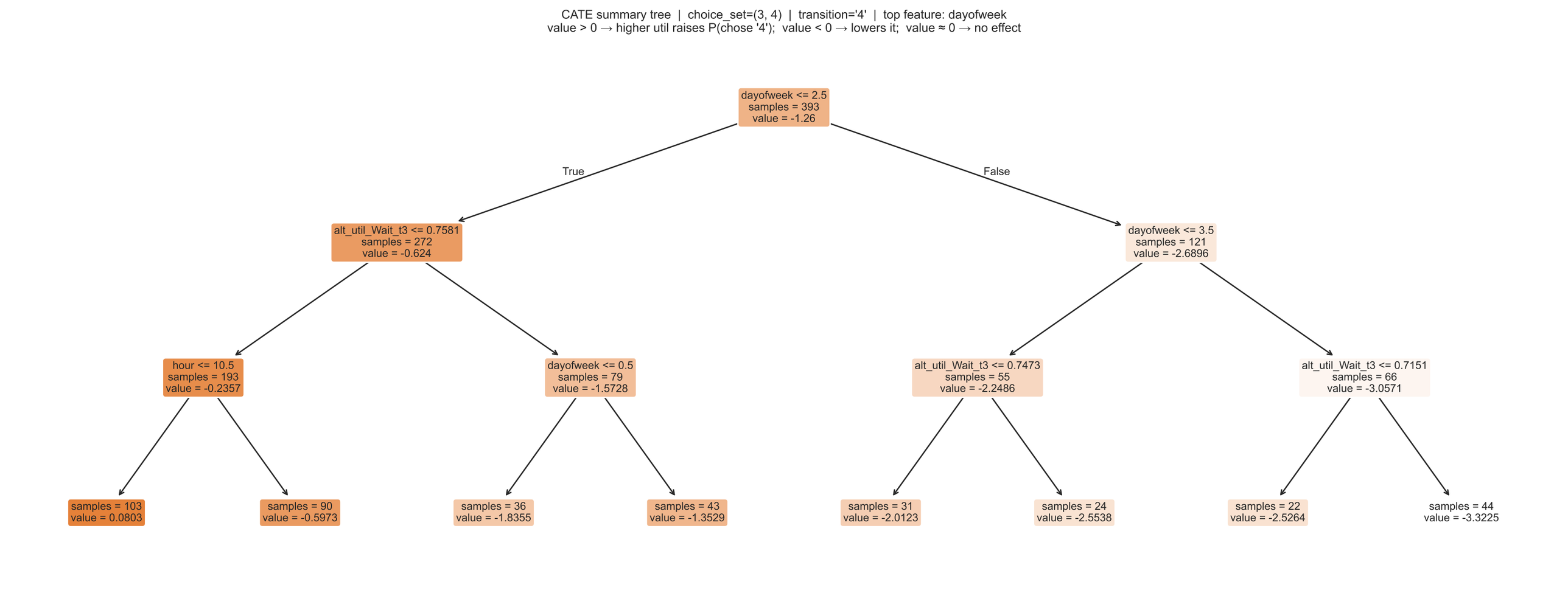
choice_set	choice_set_labels	n	choices_seen	transition	transition_label	alt_utils	ate	ci_95	p_value	reject_H0	note
(0, 1, 2)	('т_t0', 'т_t1', 'Unmatched_t2')	4	2	—	—	—	nan	—	nan	—	too few observations
(3, 4)	('Wait_t3', 'Assigned_t4')	393	3, 4	3	Wait_t3	Assigned_t4	-2.18279	[-5.0498, 0.6842]	0.1356	False	ATE=-2.1828: 0→1 util ↓ P(t3/Wait_t3) by 218.3%
(3, 4)	('Wait_t3', 'Assigned_t4')	393	3, 4	4	Assigned_t4	Wait_t3	-1.259997	[-4.2909, 1.7709]	0.4152	False	ATE=-1.2600: 0→1 util ↓ P(t4/Assigned_t4) by 126.0%
(5, 6, 7, 8)	('т_t5', 'т_t6', 'In Progress_t7', 'Awaiting Assignment_t8')	203	7, 8	7	In Progress_t7	Awaiting Assignment_t8	0.07268	[-3.1787, 3.3241]	0.9651	False	ATE=+0.0727: 0→1 util ↑ P(t7/In Progress_t7) by 7.3%
(5, 6, 7, 8)	('т_t5', 'т_t6', 'In Progress_t7', 'Awaiting Assignment_t8')	203	7, 8	8	Awaiting Assignment_t8	In Progress_t7	-0.102648	[-3.1540, 2.9487]	0.9474	False	ATE=-0.1026: 0→1 util ↓ P(t8/Awaiting Assignment_t8) by 10.2
(6, 7, 8)	('т_t6', 'In Progress_t7', 'Awaiting Assignment_t8')	1243	7, 8	7	In Progress_t7	Awaiting Assignment_t8	-0.205229	[-1.8880, 1.4776]	0.8111	False	ATE=-0.2052: 0→1 util ↓ P(t7/In Progress_t7) by 20.5%
(6, 7, 8)	('т_t6', 'In Progress_t7', 'Awaiting Assignment_t8')	1243	7, 8	8	Awaiting Assignment_t8	In Progress_t7	-0.93887	[-2.8727, 0.9950]	0.3413	False	ATE=-0.9389: 0→1 util ↓ P(t8/Awaiting Assignment_t8) by 93.9
(7, 8)	('In Progress_t7', 'Awaiting Assignment_t8')	2239	7, 8	7	In Progress_t7	Awaiting Assignment_t8	-0.074956	[-3.5606, 3.4107]	0.9664	False	ATE=-0.0750: 0→1 util ↓ P(t7/In Progress_t7) by 7.5%
(7, 8)	('In Progress_t7', 'Awaiting Assignment_t8')	2239	7, 8	8	Awaiting Assignment_t8	In Progress_t7	0.1597	[-4.0013, 4.3207]	0.94	False	ATE=+0.1597: 0→1 util ↑ P(t8/Awaiting Assignment_t8) by 16.
(9, 10)	('т_t9', 'Closed_t10')	1446	10	—	—	—	nan	—	nan	—	too few observations



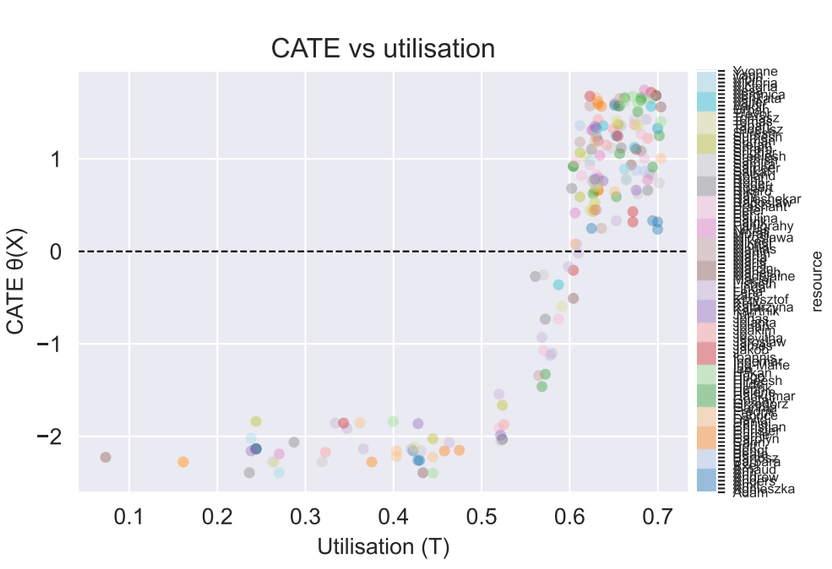

CATE — choice_set=(3, 4) | transition='3' | n=393 | ATE=-2.1828 std=1.0355
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=-2.1828
Count
CATE θ(X)
Mean CATE by resource
Tomas
Roland
Robert
Per
Miroslawa
Mikael
Lisbeth
Linda
Katarzyna
Jonas
Jayalakshmi
Ian
Henrik
Gunilla
Fredrik
David
Damian
Andrew
Mean CATE

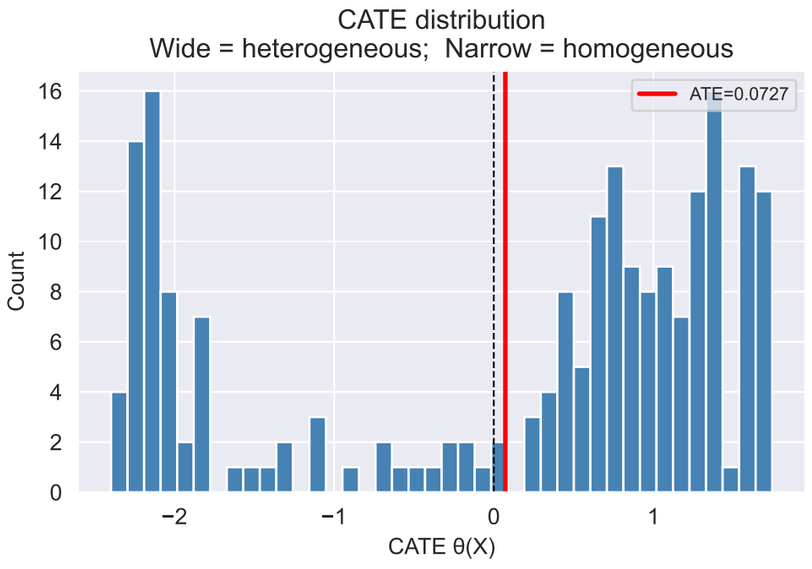

CATE summary tree | choice_set=(3, 4) | transition='3' | top feature: hour
value > 0 → higher util raises P(chose '3'); value < 0 → lowers it; value ≈ 0 → no effect
hour <= 10.5
samples = 393
value = -2.1828
True
False
hour <= 7.5
samples = 192
value = -1.7333
hour <= 12.5
samples = 201
value = -2.6122
alt_util_Assigned_t4 <= 0.6745
samples = 63
value = -2.2995
alt_util_Assigned_t4 <= 0.7123
samples = 129
value = -1.4567
alt_util_Assigned_t4 <= 0.7028
samples = 86
value = -3.5506
hour <= 13.5
samples = 115
value = -1.9104
samples = 21
value = -1.9077
samples = 42
value = -2.4953
samples = 71
value = -1.0644
samples = 58
value = -1.937
samples = 39
value = -2.9654
samples = 47
value = -4.0362
samples = 39
value = -2.4775
samples = 76
value = -1.6195

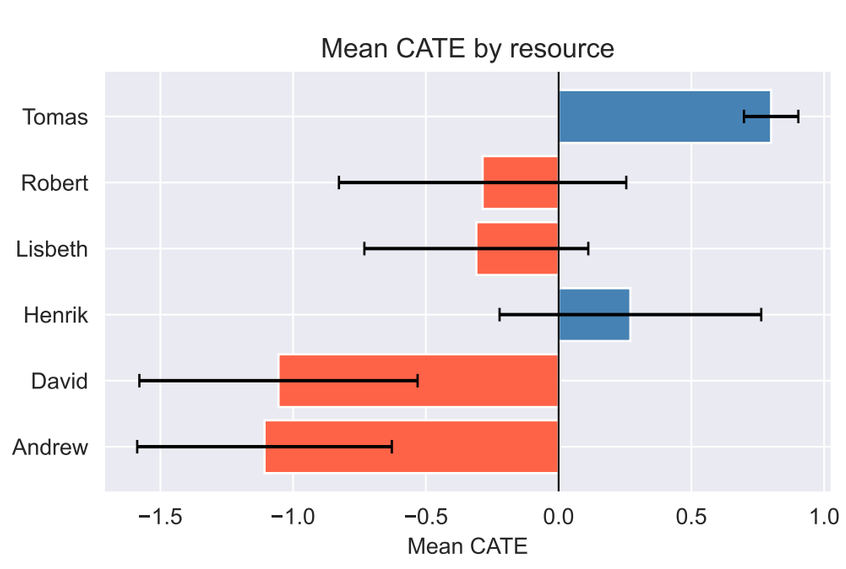

CATE — choice_set=(3, 4) | transition='4' | n=393 | ATE=-1.2600 std=1.2319
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=-1.2600
Count
CATE θ(X)
Mean CATE by resource
Tomas
Roland
Robert
Per
Miroslawa
Mikael
Lisbeth
Linda
Katarzyna
Jonas
Jayalakshmi
Ian
Henrik
Gunilla
Fredrik
David
Damian
Andrew
Mean CATE

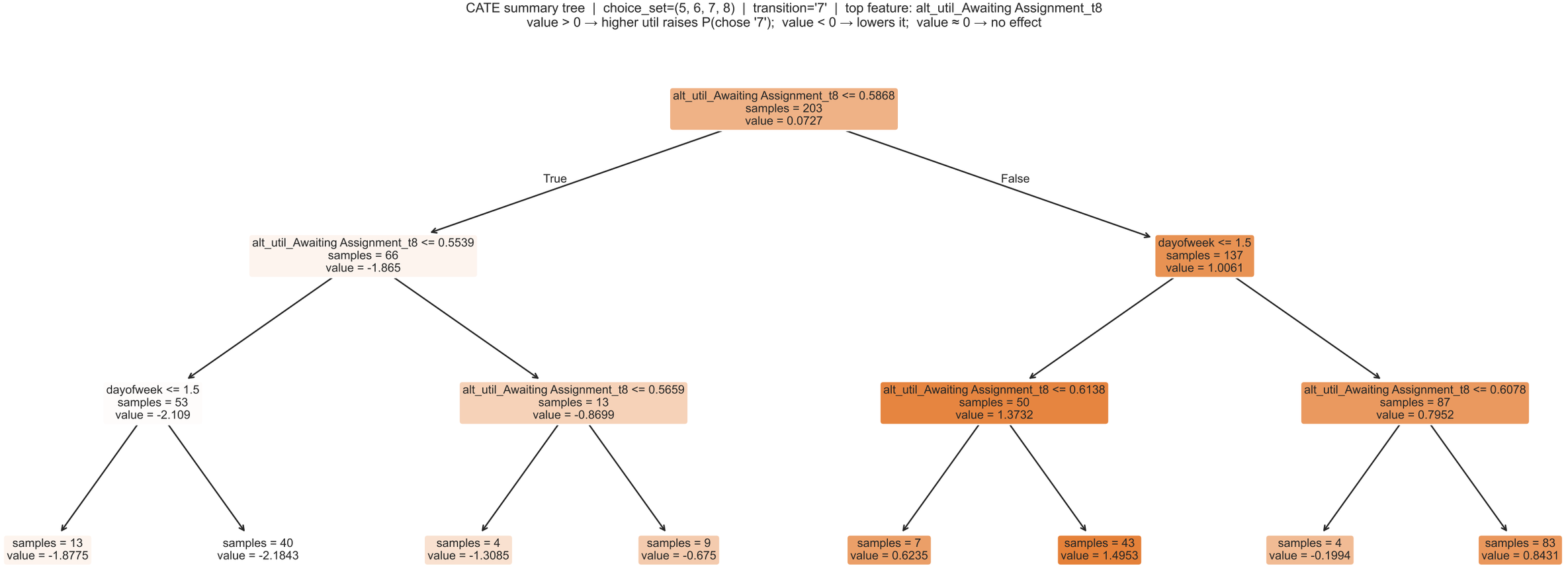

CATE summary tree | choice_set=(3, 4) | transition='4' | top feature: dayofweek
value > 0 → higher util raises P(chose '4'); value < 0 → lowers it; value ≈ 0 → no effect
dayofweek <= 2.5
samples = 393
value = -1.26
True
False
alt_util_Wait_t3 <= 0.7581
samples = 272
value = -0.624
dayofweek <= 3.5
samples = 121
value = -2.6896
hour <= 10.5
samples = 193
value = -0.2357
dayofweek <= 0.5
samples = 79
value = -1.5728
alt_util_Wait_t3 <= 0.7473
samples = 55
value = -2.2486
alt_util_Wait_t3 <= 0.7151
samples = 66
value = -3.0571
samples = 103
value = 0.0803
samples = 90
value = -0.5973
samples = 36
value = -1.8355
samples = 43
value = -1.3529
samples = 31
value = -2.0123
samples = 24
value = -2.5538
samples = 22
value = -2.5264
samples = 44
value = -3.3225

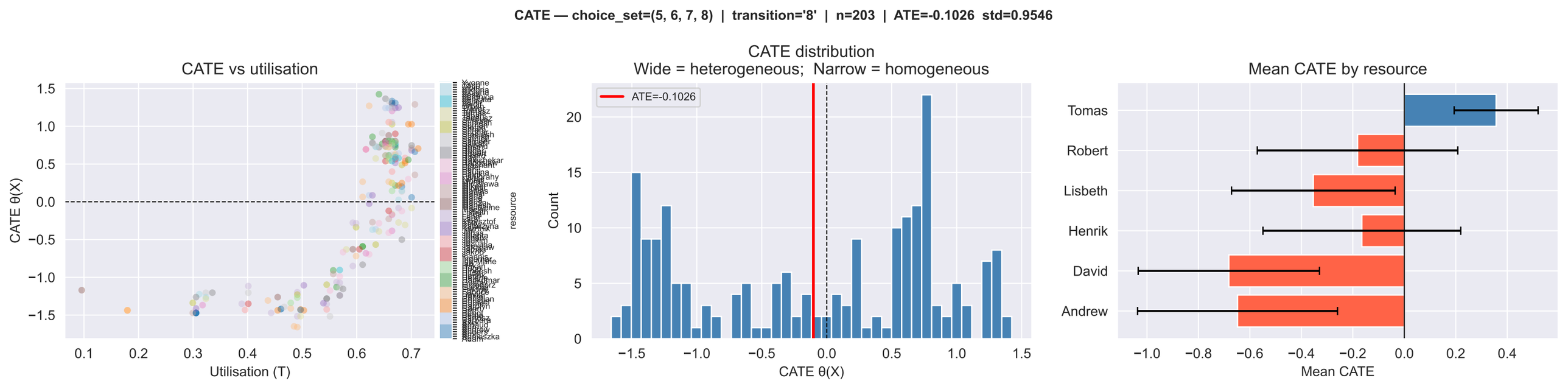
CATE — choice_set=(5, 6, 7, 8) | transition='7' | n=203 | ATE=0.0727 std=1.4350

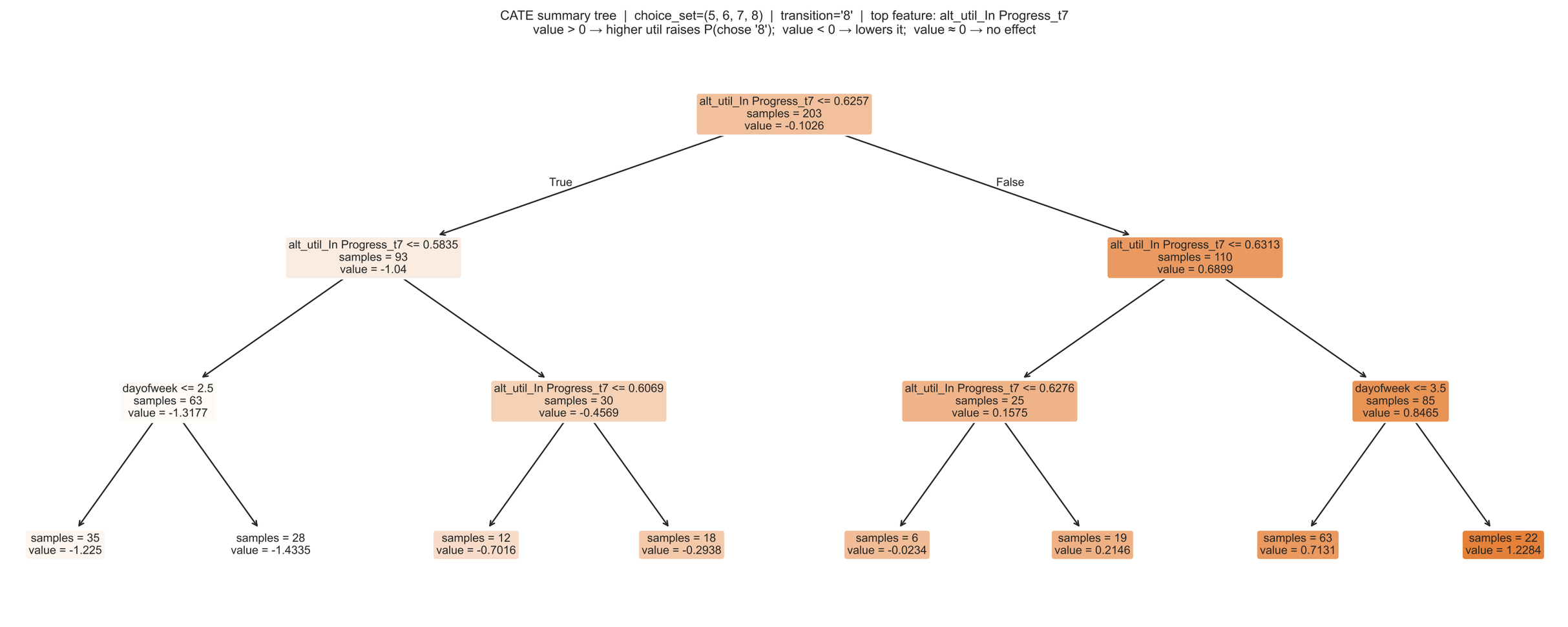


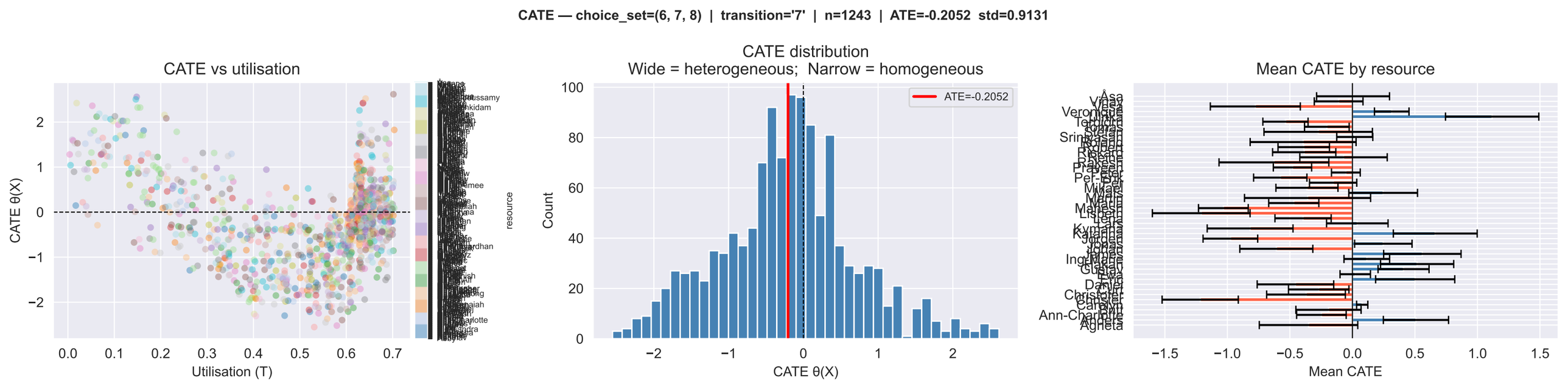


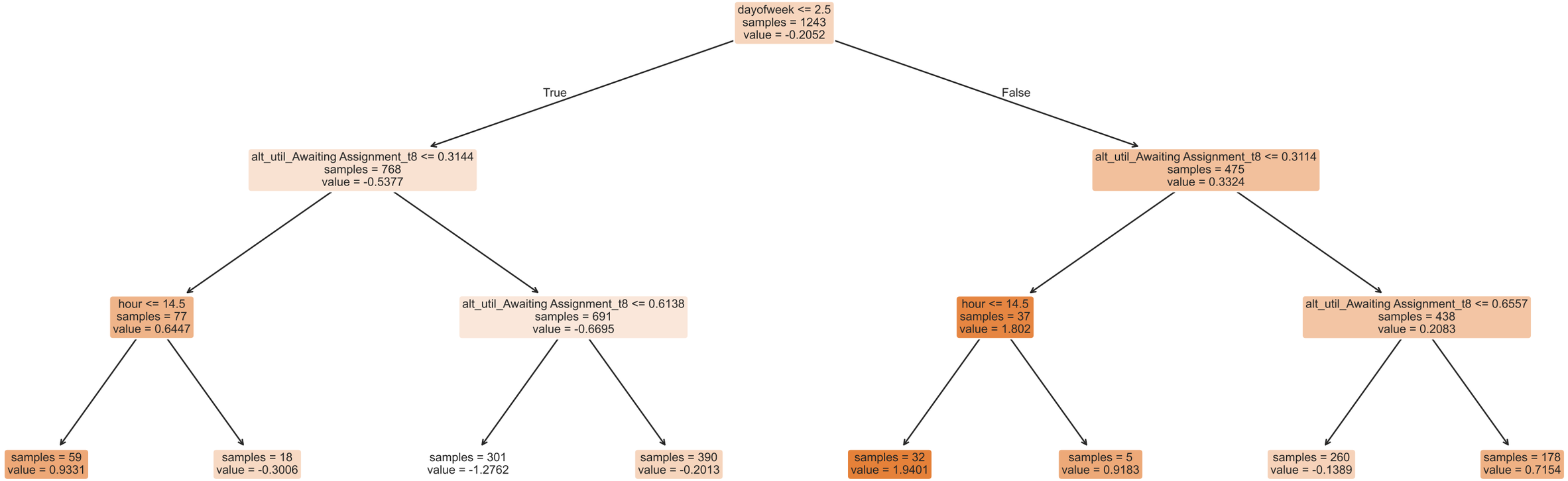


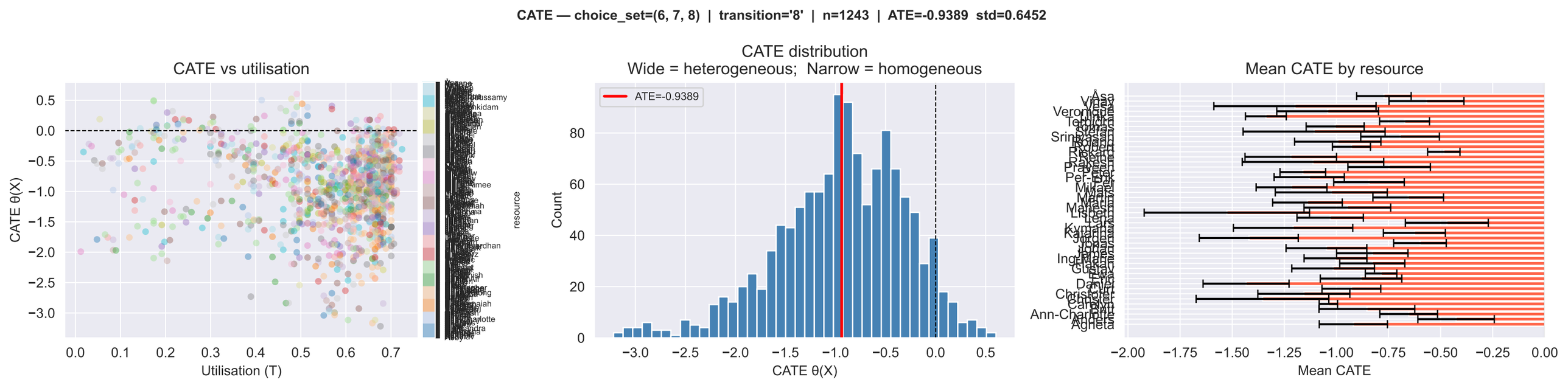

CATE summary tree | choice_set=(5, 6, 7, 8) | transition='7' | top feature: alt_util_Awaiting Assignment_t8
value > 0 → higher util raises P(chose '7'); value < 0 → lowers it; value ≈ 0 → no effect
alt_util_Awaiting Assignment_t8 <= 0.5868
samples = 203
value = 0.0727
True
False
alt_util_Awaiting Assignment_t8 <= 0.5539
samples = 66
value = -1.865
dayofweek <= 1.5
samples = 137
value = 1.0061
dayofweek <= 1.5
samples = 53
value = -2.109
alt_util_Awaiting Assignment_t8 <= 0.5659
samples = 13
value = -0.8699
alt_util_Awaiting Assignment_t8 <= 0.6138
samples = 50
value = 1.3732
alt_util_Awaiting Assignment_t8 <= 0.6078
samples = 87
value = 0.7952
samples = 13
value = -1.8775
samples = 40
value = -2.1843
samples = 4
value = -1.3085
samples = 9
value = -0.675
samples = 7
value = 0.6235
samples = 43
value = 1.4953
samples = 4
value = -0.1994
samples = 83
value = 0.8431

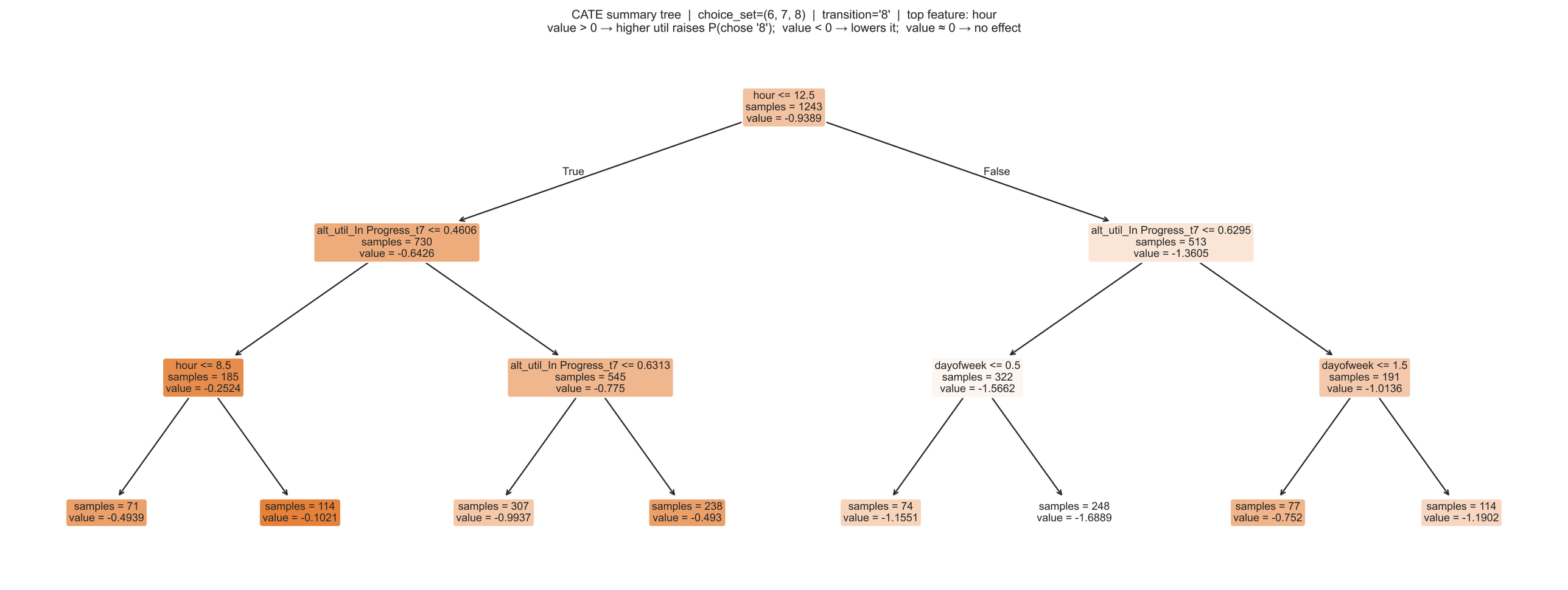

CATE — choice_set=(5, 6, 7, 8) | transition='8' | n=203 | ATE=-0.1026 std=0.9546
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=-0.1026
Count
CATE θ(X)
Mean CATE by resource
Tomas
Robert
Lisbeth
Henrik
David
Andrew
Mean CATE

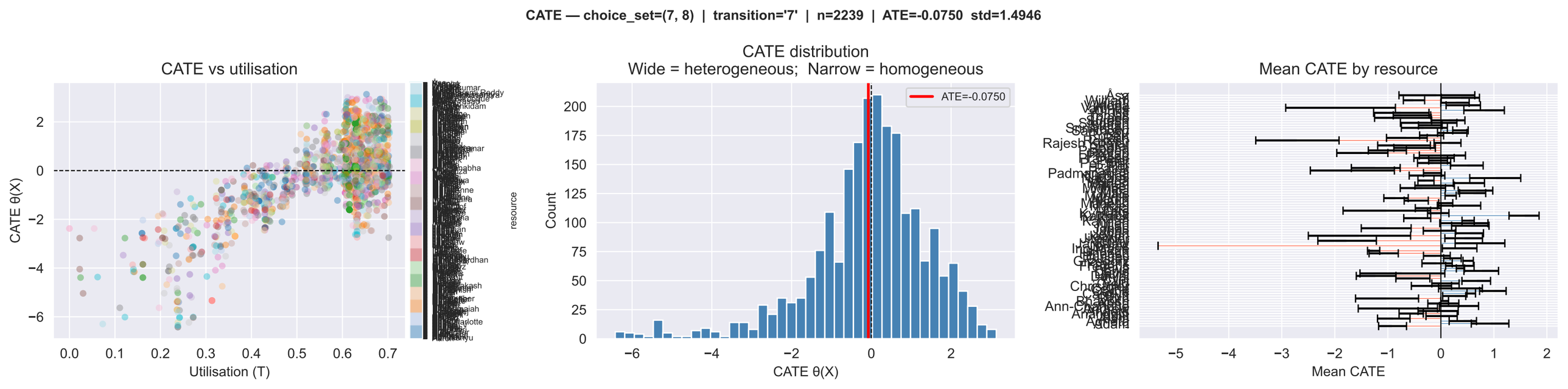

CATE summary tree | choice_set=(5, 6, 7, 8) | transition='8' | top feature: alt_util_ln Progress_t7
value > 0 → higher util raises P(chose '8'); value < 0 → lowers it; value ≈ 0 → no effect
alt_util_ln Progress_t7 <= 0.6257
samples = 203
value = -0.1026
True
False
alt_util_ln Progress_t7 <= 0.5835
samples = 93
value = -1.04
alt_util_ln Progress_t7 <= 0.6313
samples = 110
value = 0.6899
dayofweek <= 2.5
samples = 63
value = -1.3177
alt_util_ln Progress_t7 <= 0.6069
samples = 30
value = -0.4569
alt_util_ln Progress_t7 <= 0.6276
samples = 25
value = 0.1575
dayofweek <= 3.5
samples = 85
value = 0.8465
samples = 35
value = -1.225
samples = 28
value = -1.4335
samples = 12
value = -0.7016
samples = 18
value = -0.2938
samples = 6
value = -0.0234
samples = 19
value = 0.2146
samples = 63
value = 0.7131
samples = 22
value = 1.2284

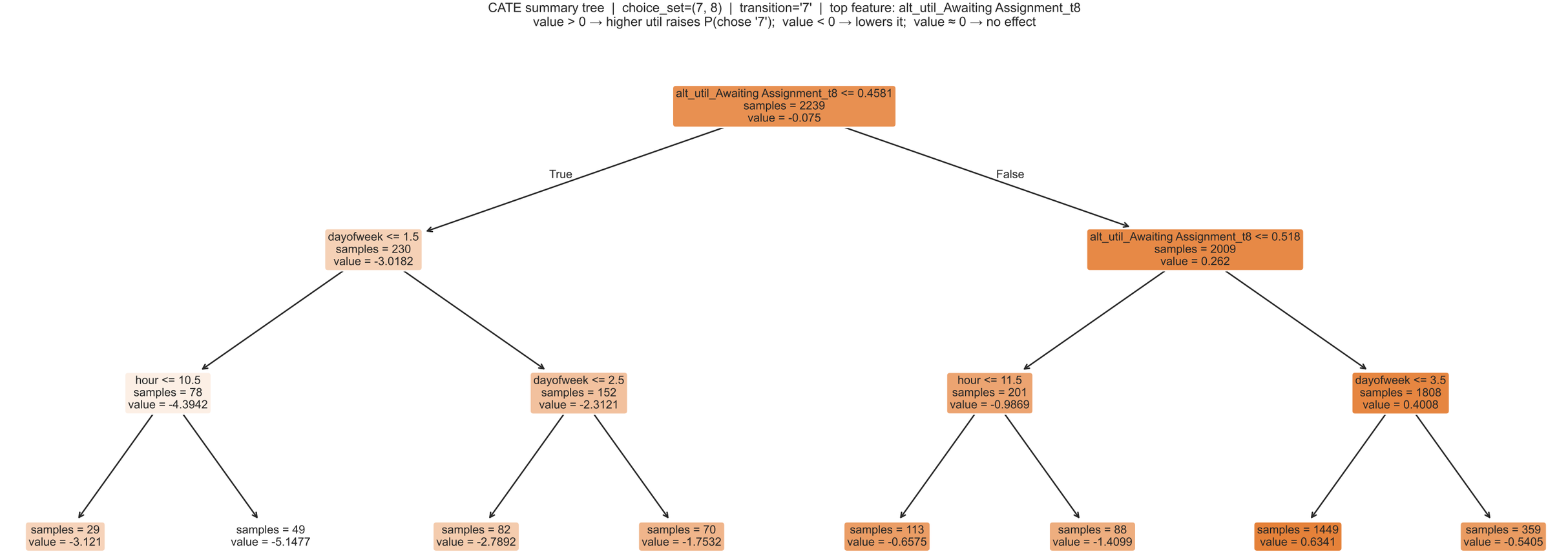

CATE — choice_set=(6, 7, 8) | transition='7' | n=1243 | ATE=-0.2052 std=0.9131
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
Count
CATE θ(X)
ATE=-0.2052
Mean CATE by resource
Mean CATE

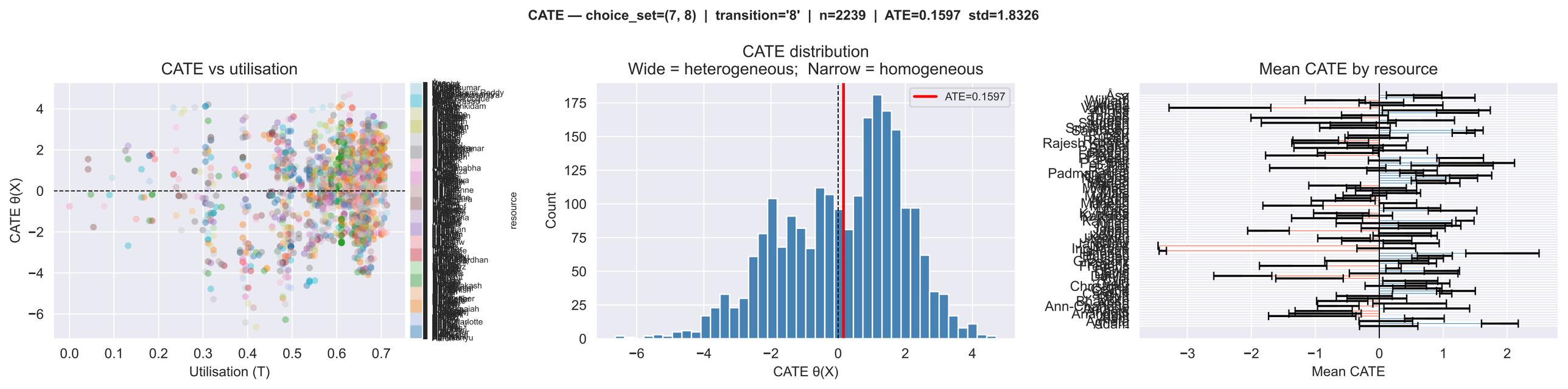

CATE summary tree | choice_set=(6, 7, 8) | transition='7' | top feature: alt_util_Awaiting Assignment_t8
value > 0 → higher util raises P(chose '7'); value < 0 → lowers it; value ≈ 0 → no effect
dayofweek <= 2.5
samples = 1243
value = -0.2052
True
False
alt_util_Awaiting Assignment_t8 <= 0.3144
samples = 768
value = -0.5377
alt_util_Awaiting Assignment_t8 <= 0.3114
samples = 475
value = 0.3324
hour <= 14.5
samples = 77
value = 0.6447
alt_util_Awaiting Assignment_t8 <= 0.6138
samples = 691
value = -0.6695
hour <= 14.5
samples = 37
value = 1.802
alt_util_Awaiting Assignment_t8 <= 0.6557
samples = 438
value = 0.2083
samples = 59
value = 0.9331
samples = 18
value = -0.3006
samples = 301
value = -1.2762
samples = 390
value = -0.2013
samples = 32
value = 1.9401
samples = 5
value = 0.9183
samples = 260
value = -0.1389
samples = 178
value = 0.7154

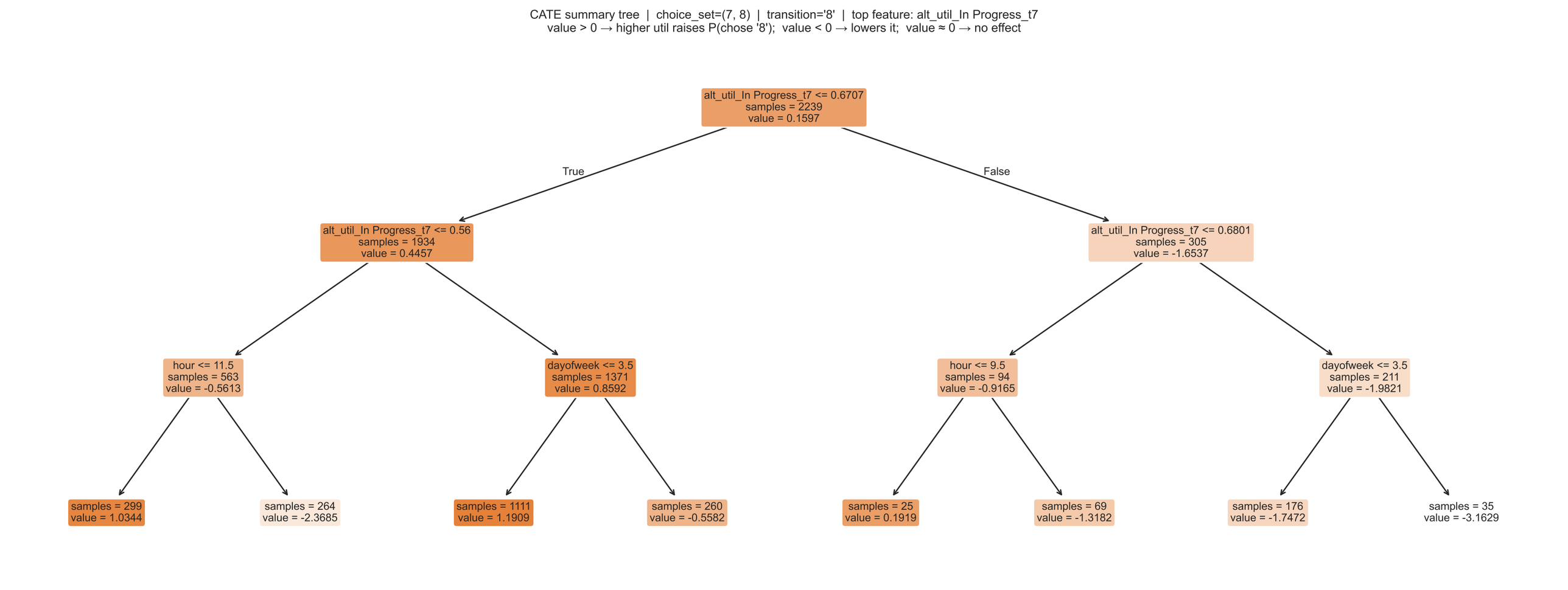

CATE — choice_set=(6, 7, 8) | transition='8' | n=1243 | ATE=-0.9389 std=0.6452
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=-0.9389
Count
CATE θ(X)
Mean CATE by resource
Mean CATE


CATE summary tree | choice_set=(6, 7, 8) | transition='8' | top feature: hour
value > 0 → higher util raises P(chose '8'); value < 0 → lowers it; value ≈ 0 → no effect
hour <= 12.5
samples = 1243
value = -0.9389
True
False
alt_util_ln Progress_t7 <= 0.4606
samples = 730
value = -0.6426
alt_util_ln Progress_t7 <= 0.6295
samples = 513
value = -1.3605
hour <= 8.5
samples = 185
value = -0.2524
alt_util_ln Progress_t7 <= 0.6313
samples = 545
value = -0.775
dayofweek <= 0.5
samples = 322
value = -1.5662
dayofweek <= 1.5
samples = 191
value = -1.0136
samples = 71
value = -0.4939
samples = 114
value = -0.1021
samples = 307
value = -0.9937
samples = 238
value = -0.493
samples = 74
value = -1.1551
samples = 248
value = -1.6889
samples = 77
value = -0.752
samples = 114
value = -1.1902


CATE — choice_set=(7, 8) | transition='7' | n=2239 | ATE=-0.0750 std=1.4946
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=-0.0750
Count
CATE θ(X)
Mean CATE by resource
Mean CATE


CATE summary tree | choice_set=(7, 8) | transition='7' | top feature: alt_util_Awaiting Assignment_t8
value > 0 → higher util raises P(chose '7'); value < 0 → lowers it; value ≈ 0 → no effect
alt_util_Awaiting Assignment_t8 <= 0.4581
samples = 2239
value = -0.075
True
False
dayofweek <= 1.5
samples = 230
value = -3.0182
alt_util_Awaiting Assignment_t8 <= 0.518
samples = 2009
value = 0.262
hour <= 10.5
samples = 78
value = -4.3942
dayofweek <= 2.5
samples = 152
value = -2.3121
hour <= 11.5
samples = 201
value = -0.9869
dayofweek <= 3.5
samples = 1808
value = 0.4008
samples = 29
value = -3.121
samples = 49
value = -5.1477
samples = 82
value = -2.7892
samples = 70
value = -1.7532
samples = 113
value = -0.6575
samples = 88
value = -1.4099
samples = 1449
value = 0.6341
samples = 359
value = -0.5405


CATE — choice_set=(7, 8) | transition='8' | n=2239 | ATE=0.1597 std=1.8326
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=0.1597
Count
CATE θ(X)
Mean CATE by resource
Mean CATE


CATE summary tree | choice_set=(7, 8) | transition='8' | top feature: alt_util_ln Progress_t7
value > 0 → higher util raises P(chose '8'); value < 0 → lowers it; value ≈ 0 → no effect
alt_util_ln Progress_t7 <= 0.6707
samples = 2239
value = 0.1597
True
False
alt_util_ln Progress_t7 <= 0.56
samples = 1934
value = 0.4457
alt_util_ln Progress_t7 <= 0.6801
samples = 305
value = -1.6537
hour <= 11.5
samples = 563
value = -0.5613
dayofweek <= 3.5
samples = 1371
value = 0.8592
hour <= 9.5
samples = 94
value = -0.9165
dayofweek <= 3.5
samples = 211
value = -1.9821
samples = 299
value = 1.0344
samples = 264
value = -2.3685
samples = 1111
value = 1.1909
samples = 260
value = -0.5582
samples = 25
value = 0.1919
samples = 69
value = -1.3182
samples = 176
value = -1.7472
samples = 35
value = -3.1629

CATE statistics summary
cate_std = heterogeneity; pct_positive/negative = share of cases where util helps/hurts; top_feature = main moderator

choice_set	transition	n	ate	cate_std	cate_min	cate_p25	cate_p50	cate_p75	cate_max	pct_positive	pct_negative	top_feature
(3, 4)	3	393	-2.18279	1.035452	-5.092358	-2.855675	-2.083146	-1.502215	0.490961	1.8	98.2	hour
(3, 4)	4	393	-1.259997	1.231891	-3.731373	-2.257869	-1.116664	-0.391419	1.637742	12.7	87.3	dayofweek
(5, 6, 7, 8)	7	203	0.07268	1.434993	-2.39763	-1.751094	0.696023	1.250741	1.744138	65.5	34.5	alt_util_Awaiting Assignment_t8
(5, 6, 7, 8)	8	203	-0.102648	0.95461	-1.658912	-1.172958	0.085722	0.733833	1.424135	52.2	47.8	alt_util_In Progress_t7
(6, 7, 8)	7	1243	-0.205229	0.913113	-2.552399	-0.739461	-0.197474	0.323349	2.6174	39.3	60.7	alt_util_Awaiting Assignment_t8
(6, 7, 8)	8	1243	-0.93887	0.645165	-3.218108	-1.323548	-0.89874	-0.467956	0.603136	5.1	94.9	hour
(7, 8)	7	2239	-0.074956	1.494599	-6.422404	-0.683293	0.063754	0.793367	3.134199	52.8	47.2	alt_util_Awaiting Assignment_t8
(7, 8)	8	2239	0.1597	1.832554	-6.638585	-1.210992	0.532981	1.528826	4.711367	57.8	42.2	alt_util_In Progress_t7

Key Assumptions & Interpretation Notes

Backdoor criterion (all methods)

X = {hour, dayofweek, n_alternatives, res_*} must block all confounding paths T ← C → Y.

Any variable that affects both util and routing must be in X. Unobserved confounders

cannot be detected and will bias all estimates.

SUTVA (all methods)

One case's routing does not affect another's treatment or outcome.

Violated if resource occupation by one case changes the util seen by a concurrent case.

Positivity (all methods)

Every covariate combination X must have positive probability for all util levels T.

Diagnosed by the covariate balance (SMD) in the backdoor check.

Linearity (Double ML only)

θ is the best-fit linear slope. If the effect varies with X, θ is a variance-weighted

average of CATEs — valid as a summary, but not the simple mean.

Double ML is more powerful than the forest for testing 'is there any effect?'

Honest splitting (Causal Forest only)

Tree structure and leaf estimates use separate subsamples. Yields valid CIs but

halves effective sample size → forest CIs are wider than Double ML's.

A significant Double ML θ with an insignificant forest ATE is not a contradiction.

What the analysis cannot establish

Causal direction (assumes util → choice, not the reverse).

External validity beyond the observed covariate distribution.

Dynamic/lagged effects across cases over time.

Full documentation: ASSUMPTIONS.md